

# Lecture IX - Data Visualization

## Programming with Python

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### EP 9: The Pitch Deck

#### Charts are arguments

**Friday**, the boardroom. Picture it: the investor slides a pencil out of her jacket and **sharpens it** while you set up, the sound louder than it has any right to be.

...

Last week you handed her a table she could trust. She read exactly one row, looked up, and said the thing this whole session is about:

**“Tables are homework. Charts are arguments.”**

...

A number convinces the careful. A **picture** convinces the room. Today: draw the right chart for the question, and refuse to draw a misleading one.

#### Warm-up

##### Question 1

Three questions from Episode 8. Commit. Hands up **before** the reveal.

```
orders[orders["zone"] == "Nord"]
```

returns...

a) a True/False column b) only the Nord rows c) an error from comparing text

##### Answer 1

```
orders[orders["zone"] == "Nord"]
```

|   | order_id | zone | items | total_eur |
|---|----------|------|-------|-----------|
| 0 | 101      | Nord | 2     | 18.5      |
| 2 | 103      | Nord | 3     | 24.0      |

...

b) **only the Nord rows**: the True/False column is what the **inner** expression `orders["zone"] == "Nord"` makes. Wrapped in `orders[...]`, that mask keeps the rows where it's **True** and drops the rest.

## Question 2

Tobi's AI wrote this. `orders` is a normal DataFrame. What happens?

```
orders.summarize()
```

a) a summary table b) an empty DataFrame c) an `AttributeError`

## Answer 2

```
orders.summarize()
```

```
AttributeError: 'DataFrame' object has no attribute 'summarize'
[31mAttributeError[39m[31m: [39m 'DataFrame' object has no attribute
'summarize'
```

...

c) **AttributeError**: pandas has `.describe()`, not `.summarize()`. The AI invented a plausible name. An AI that sounds sure is not the same as an API that exists. You verify, every time.

## Question 3

```
orders["total_eur"].describe()
```

shows...

a) the first five rows of the column, like `.head()` b) count / mean / std / min / quartiles / max c) the column's dtype, like `.dtypes`

## Answer 3

```
orders["total_eur"].describe()
```

```
count      5.000000
mean       17.580000
std        10.688873
min         6.800000
25%         7.200000
50%        18.500000
75%        24.000000
max        31.400000
Name: total_eur, dtype: float64
```

...

b) **count / mean / std / min / quartiles / max**. One line, the column's whole statistical fingerprint. That's the real method Tobi's AI was reaching for.

## The first chart

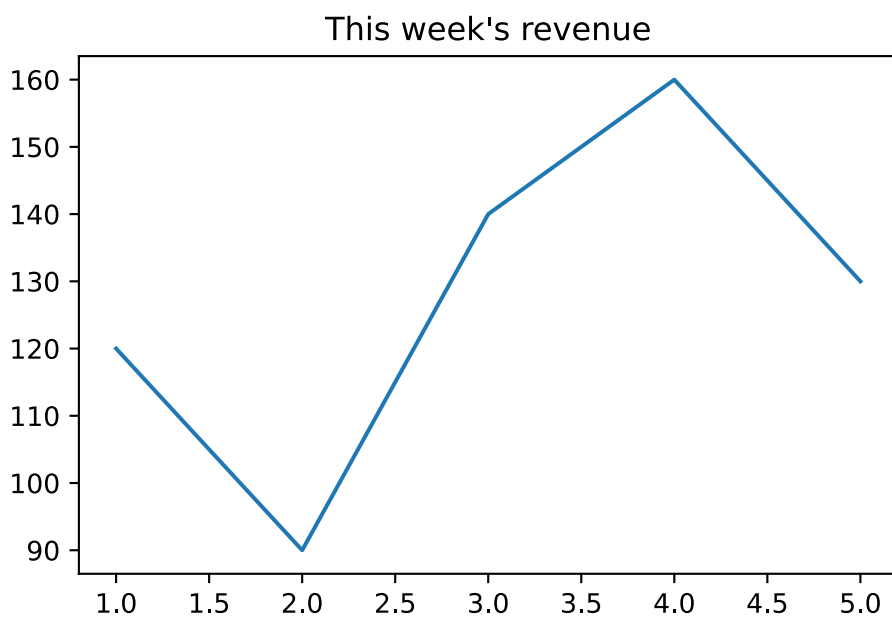
### Matplotlib in four lines

Everyone plots with **matplotlib**, imported as `plt`. Four lines turn a list of numbers into a picture:

```
import matplotlib.pyplot as plt

days = [1, 2, 3, 4, 5]
revenue = [120, 90, 140, 160, 130]

plt.figure()          # OPEN: a fresh canvas, so this chart
                        #      doesn't draw on the last one
plt.plot(days, revenue) # x, then y
plt.title("This week's revenue")
plt.gca()              # SHOW: "get current axes" (the chart)
```



...

### Note

Two habits for every chart cell: open with `plt.figure()` (a clean canvas) and end with `plt.gca()`. In marimo there's no `plt.show()`; the figure appears as the cell's **last expression**, and `plt.plot(...)` alone returns line objects, not a picture. Today's lab is a script and has no cell that could show a chart, so there a chart ends in a picture file. More on that before the break.

## Predict: what's on the x-axis?

Tobi is in a hurry and hands `plot` **one** list, no `days`:

```
plt.plot([3, 5, 4])
```

What does the x-axis show?

a) 3, 5, 4, the values themselves b) 0, 1, 2, the list positions c) 1, 2, 3, counted the human way

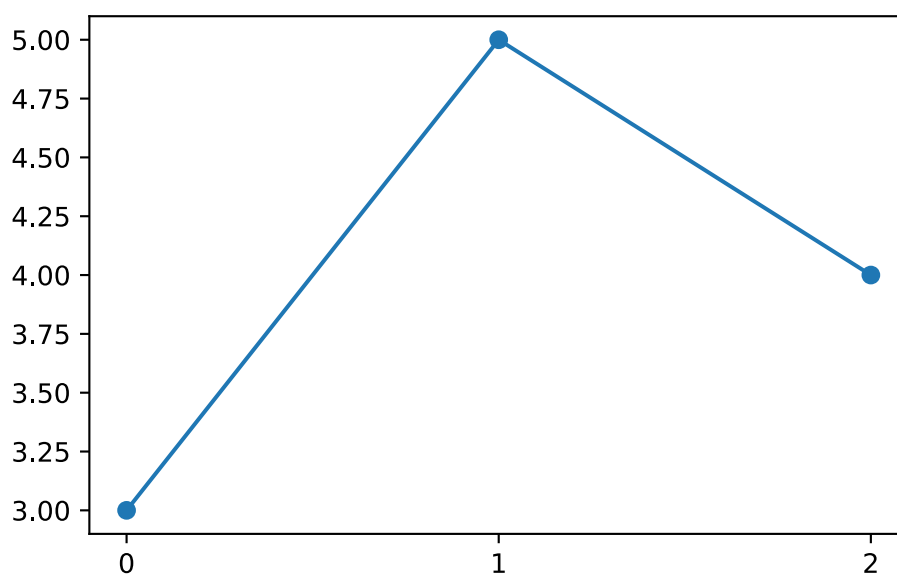
...

Predict. Pick a letter, then I reveal the answer.

## Answer: the positions, from zero

```
import matplotlib.pyplot as plt

plt.figure()
plt.plot([3, 5, 4], marker="o")
plt.xticks([0, 1, 2])  # show only whole positions
plt.gca()
```



...

b) 0, 1, 2: with one list, matplotlib takes it as **y** and invents x as the positions, and Python counts positions from 0.

...

Want days 1-5 on the axis? Pass them: `plt.plot(days, revenue)`, x first.

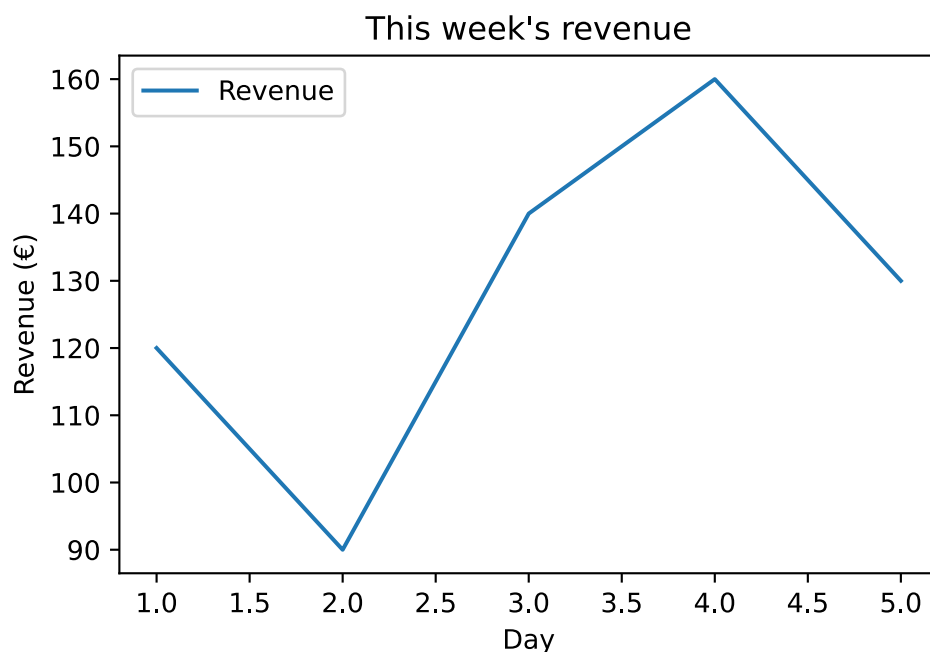
## The parts of a chart

Every chart names its axes. `xlabel`, `ylabel`, `title`, and (when there's more than one line) a `legend`:

```
import matplotlib.pyplot as plt

days = [1, 2, 3, 4, 5]
revenue = [120, 90, 140, 160, 130]

plt.figure()
plt.plot(days, revenue, label="Revenue") # label feeds the legend
plt.xlabel("Day")                       # what the x-axis counts
plt.ylabel("Revenue (€)")               # what the y-axis measures
plt.title("This week's revenue")        # what the picture is about
plt.legend()                            # the little key, one entry per label
plt.gca()
```



## Your turn

### Open the exercise

Scan the QR or type the link:

[python.tobiasvlcek.com/notebooks/ex\\_09\\_a/](https://python.tobiasvlcek.com/notebooks/ex_09_a/)



**Any questions?** Let me know!

### A little styling: and only a little

Two keywords cover almost everything: `color` and `linestyle`. Restraint is the professional move. One clear line beats a rainbow:

```
import matplotlib.pyplot as plt

days = [1, 2, 3, 4, 5]
revenue = [120, 90, 140, 160, 130]

plt.figure()
plt.plot(days, revenue, color="crimson", linestyle="--")
plt.title("Styled, not decorated")
plt.gca()
```



### Predict: one chart or two?

Tobi runs **two** `plot` calls, back to back, with no `plt.figure()` between them:

```
plt.plot([3, 1, 2])  
plt.plot([1, 2, 3])
```

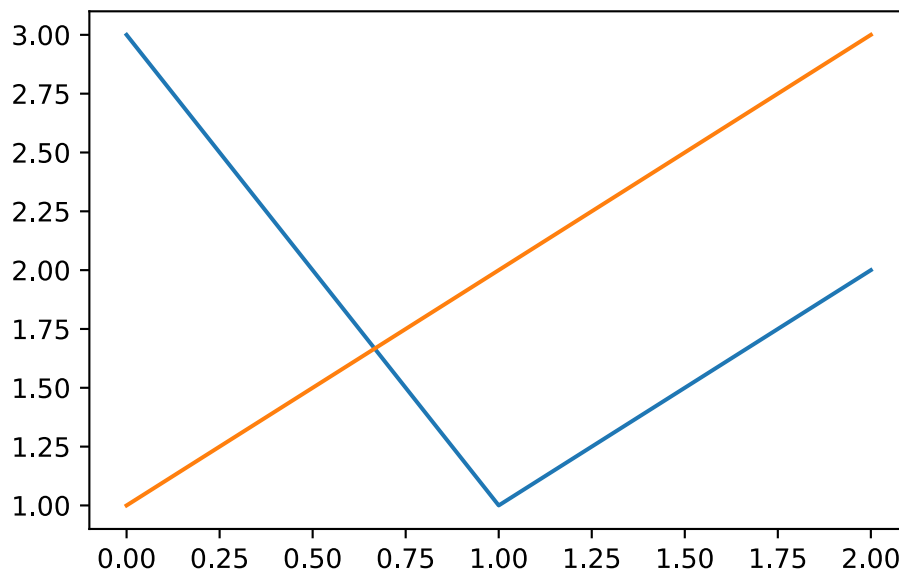
a) one chart with two lines b) two separate charts, one per call c) an error: the figure is already in use

...

Predict. Pick a letter, then I reveal the answer.

### Answer: one chart, two lines

```
import matplotlib.pyplot as plt  
  
plt.figure()  
plt.plot([3, 1, 2])  
plt.plot([1, 2, 3])    # same canvas: a second line, not a second chart  
plt.gca()
```



...

a) **one chart with two lines**, because matplotlib keeps drawing on the **current** figure until you start a new one with `plt.figure()`.

...

That's exactly why every chart cell **opens** with `plt.figure()`: it's how you say "new picture, start clean."

## Your turn

### Open the exercise

Scan the QR or type the link:

[python.tobiasvlcek.com/notebooks/ex\\_09\\_b/](https://python.tobiasvlcek.com/notebooks/ex_09_b/)



Any questions? Let me know!

## The right chart

### Match the chart to the question

The wrong chart is its own kind of lie. Start from the **question**, not the chart:

| The question                  | The chart | The call                              |
|-------------------------------|-----------|---------------------------------------|
| Compare <b>categories</b> ?   | bar       | <code>plt.bar(labels, heights)</code> |
| See a <b>distribution</b> ?   | histogram | <code>plt.hist(values)</code>         |
| <b>Two numbers</b> per order? | scatter   | <code>plt.scatter(x, y)</code>        |
| Change over <b>time</b> ?     | line      | <code>plt.plot(x, y)</code>           |

...

#### Note

**Line** you already own from block one. Missing on purpose: the **pie chart**. The eye can't compare slice sizes; a bar is clearer.

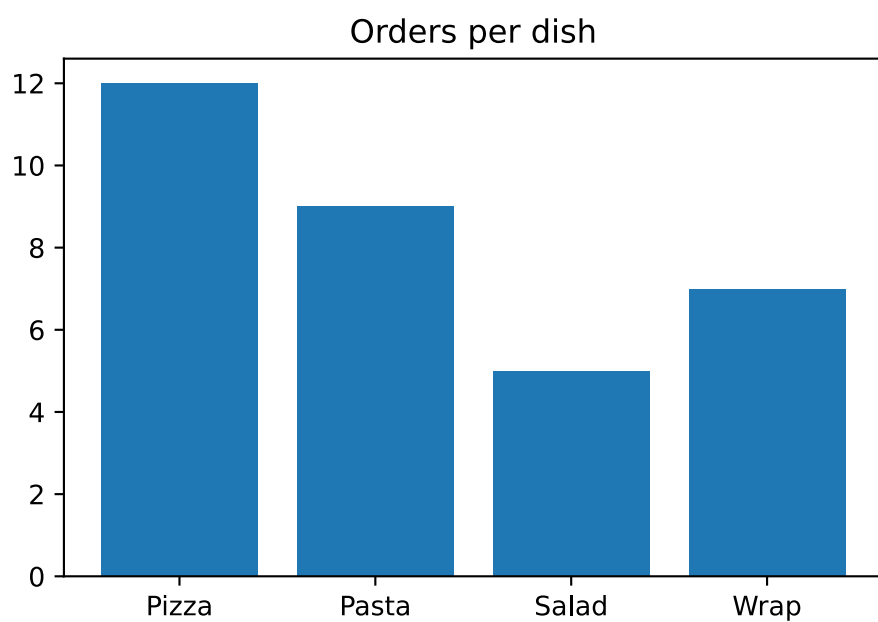
### Categories → bar

“Which dish sells best?” Categories to compare, so that's a **bar**:

```
import matplotlib.pyplot as plt

dishes = ["Pizza", "Pasta", "Salad", "Wrap"]
sold = [12, 9, 5, 7]

plt.figure()
plt.bar(dishes, sold)
plt.title("Orders per dish")
plt.gca()
```



The tallest bar answers the question at a glance: Pizza.

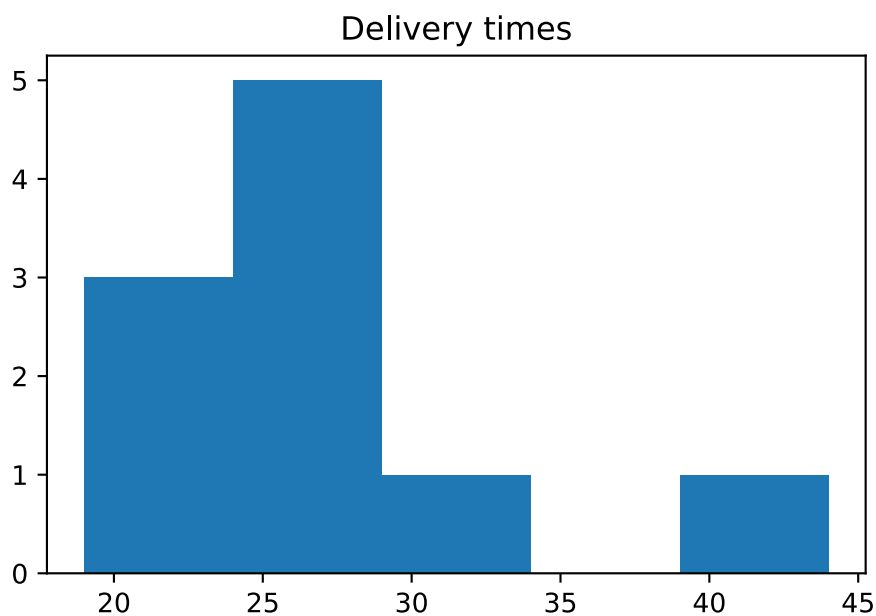
### A distribution → histogram

*"How are our delivery times spread out?"* One column of numbers dropped into buckets. That's a **histogram**:

```
import matplotlib.pyplot as plt

minutes = [19, 22, 23, 24, 24, 25, 26, 28, 31, 44]

plt.figure()
plt.hist(minutes, bins=5)
plt.title("Delivery times")
plt.gca()
```



Most deliveries land in the mid-20s, with one lonely slow one far right. A histogram shows **shape**, not individual values.

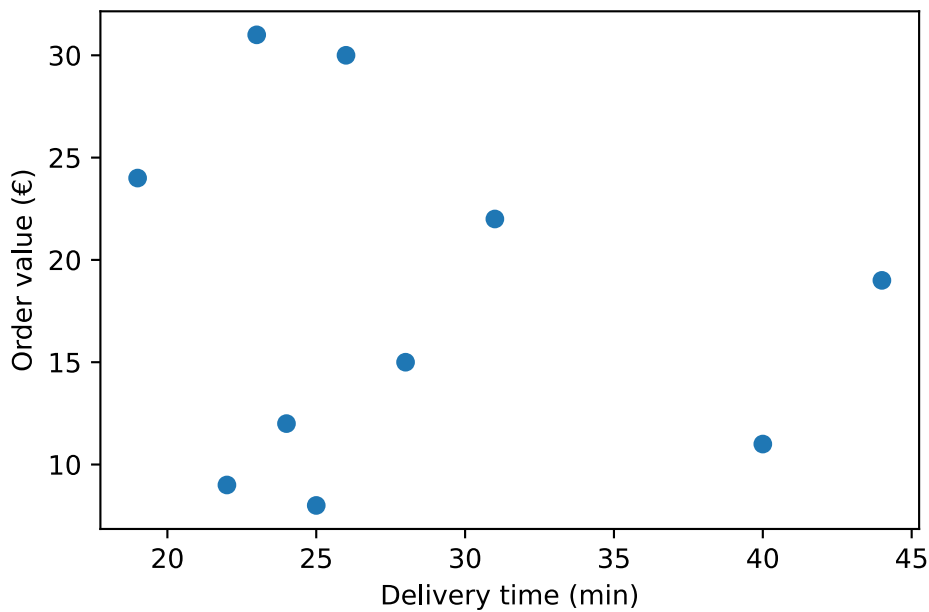
### Two numbers → scatter

*“Do bigger orders take longer?”* Two numbers per order, one on each axis, which makes it a **scatter**:

```
import matplotlib.pyplot as plt

minutes = [19, 22, 23, 24, 25, 26, 28, 31, 40, 44]
euros   = [24, 9, 31, 12, 8, 30, 15, 22, 11, 19]

plt.figure()
plt.scatter(minutes, euros)
plt.xlabel("Delivery time (min)")
plt.ylabel("Order value (€)")
plt.gca()
```



Just a cloud. No upward drift. Sometimes the answer is “**there’s no pattern here.**”

### **Predict: plot instead of scatter**

Tobi types `plot` where he meant `scatter`. Five orders, in the order they came in:

```
minutes = [31, 19, 44, 22, 28]
euros   = [22, 24, 19, 9, 15]
plt.plot(minutes, euros)
```

a) a zig-zag line hopping between the points in list order b) the same dots as `scatter`, joined into a smooth trend c) an error: `plot` refuses x values that are not sorted ascending

...

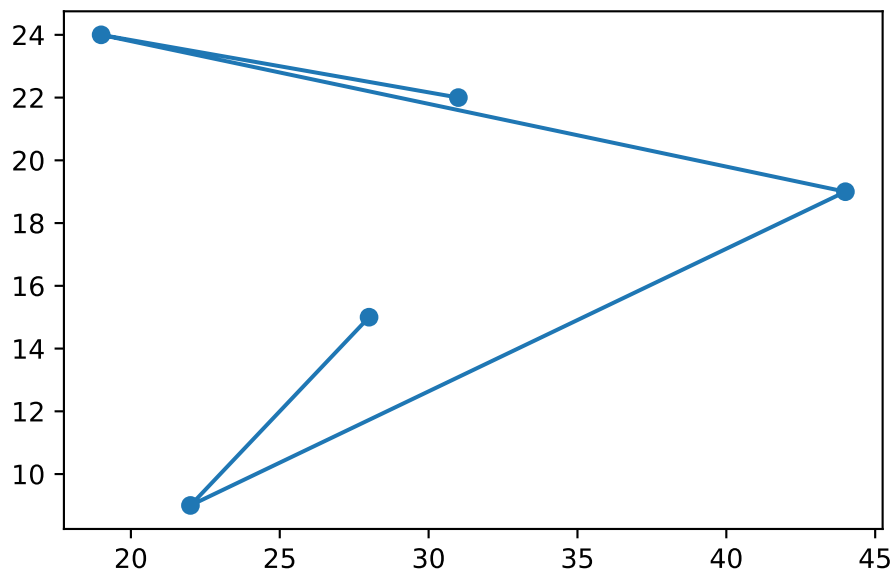
Predict. Pick a letter, then I reveal the answer.

### **Answer: a zig-zag in list order**

```
import matplotlib.pyplot as plt

minutes = [31, 19, 44, 22, 28]
euros   = [22, 24, 19, 9, 15]

plt.figure()
plt.plot(minutes, euros, marker="o") # joined in LIST order
plt.gca()
```



...

a) a **zig-zag**: `plot` connects the points **in the order you gave them**, never mind the x values. Order 1 to order 2 to order 3, back and forth.

...

A line says the points follow each other, and orders don't. Two numbers per order with no sequence between them is a job for **scatter**.

### Predict: how many bars?

The **same four numbers**, two ways. On the left, a bar per number; on the right, a histogram:

```
values = [8, 12, 9, 15]
plt.bar(range(4), values) # left
plt.hist(values, bins=3)  # right
```

The bar chart shows 4 bars. How many bars does the **histogram** show?

a) 4: one bar per number b) 15: one bar per unit up to the max c) about 2-3 lumps: it counts ranges

...

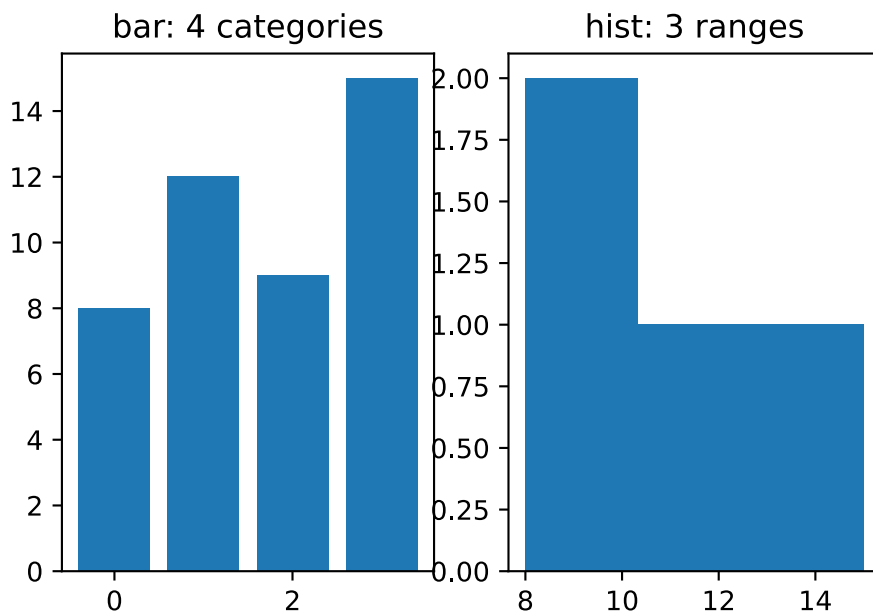
Predict. Pick a letter, then I reveal the answer.

### Answer: about 2-3 lumps

```
import matplotlib.pyplot as plt

values = [8, 12, 9, 15]
```

```
plt.figure()
plt.subplot(1, 2, 1)
plt.bar(range(4), values)      # one bar per NUMBER
plt.title("bar: 4 categories")
plt.subplot(1, 2, 2)
plt.hist(values, bins=3)      # counts how many per RANGE
plt.title("hist: 3 ranges")
plt.gca()
```



...

**c) about 2-3 lumps:** the histogram groups the numbers into ranges and counts how many fall in each. 8 and 9 land in the same bucket, so that bar is two tall.

...

**Bar counts CATEGORIES; histogram counts RANGES.** Same numbers, different question.

## Your turn

### Open the exercise

Scan the QR or type the link:

[python.tobiasvlcek.com/notebooks/ex\\_09\\_c/](https://python.tobiasvlcek.com/notebooks/ex_09_c/)



**Any questions?** Let me know!



Tip

A five-minute pause after this one, then the axis and the AI chart assistant.

## The honest axis

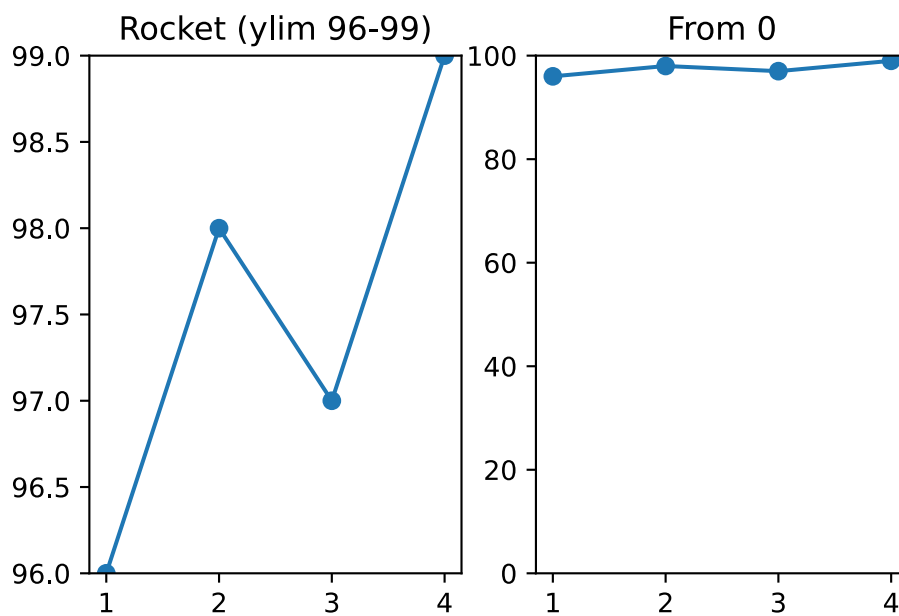
### The same data, told two ways

Four weeks of revenue, [96, 98, 97, 99], barely moving. **Where the y-axis starts** decides: the left gets you a term sheet, the right gets you trusted. Same numbers, and the room sees a rocket or the truth:

```
import matplotlib.pyplot as plt

weeks = [1, 2, 3, 4]
revenue = [96, 98, 97, 99]

plt.figure()
plt.subplot(1, 2, 1)
plt.plot(weeks, revenue, marker="o")
plt.ylim(96, 99)          # tight: DRAMA
plt.title("Rocket (ylim 96-99)")
plt.subplot(1, 2, 2)
plt.plot(weeks, revenue, marker="o")
plt.ylim(0, 100)          # from zero
plt.title("From 0")
plt.gca()
```



### Predict: the AI's one extra line

Revenue [710, 718, 724, 731], up 3 % over four weeks. Tobi's AI drafts the chart and adds one line:

```
plt.plot([1, 2, 3, 4], [710, 718, 724, 731])
plt.ylim(700, 735)
```

What does the room see?

- a) a flat line: 3 % stays 3 % on any axis
- b) an error: `ylim` has to include zero on a line chart
- c) a line climbing almost the whole chart height

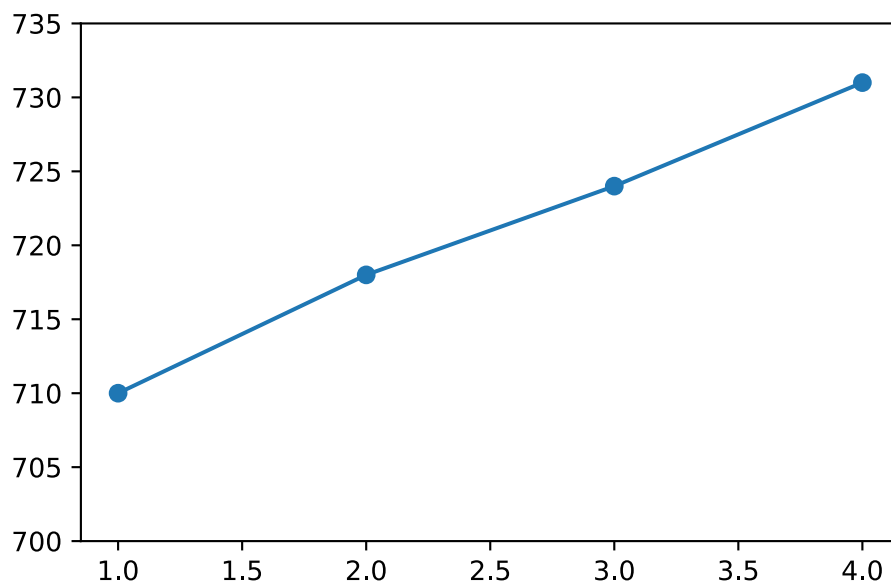
...

Predict. Pick a letter, then I reveal the answer.

### Answer: a climb across the chart

```
import matplotlib.pyplot as plt

plt.figure()
plt.plot([1, 2, 3, 4], [710, 718, 724, 731], marker="o")
plt.ylim(700, 735) # 3 % of growth, 60 % of the chart
plt.gca()
```



...

**c) a climb across the chart:** the axis spans 35 units and the line rises 21 of them, **60 % of the picture** for 3 % of growth.

...

Python raises no error and no warning here. The code is fine; the axis is what misleads.

### The rule: growth starts at zero

- A **growth claim** on a chart starts the y-axis at **0**. Anything else magnifies a wobble into a cliff.
- The investor **will** check: she reads the axis before she reads the line.

...

#### 💡 Tip

She's the type who reads Formular 27b/6 for fun. A stretched axis is the first thing she catches, and the last thing you want to explain.

## Your turn

### Open the exercise

Scan the QR or type the link:

[python.tobiasvlcek.com/notebooks/ex\\_09\\_d/](https://python.tobiasvlcek.com/notebooks/ex_09_d/)



**Any questions?** Let me know!

## AI as your chart assistant

AI is good at plotting code, if you check the result:

1. **Describe** the data and the question: “I have `orders` with columns `zone` (text) and `total_eur` (float). Bar chart of total revenue per zone.”
2. Let it **draft** the plot code.
3. **Verify** three things before you believe the picture:
  - Do the **columns** it used actually exist?
  - Does the **y-axis** start where you claim?
  - Does the **chart type** fit the question?

...

AI drafts. You decide what goes on the slide.

## Two ways AI charts lie

- **Invented column names.** The AI writes `orders["revenue"]`, but your column is `total_eur`: a `KeyError`, exactly the confident-nonsense you caught in Episode 8. Read the code before you run it.
- **Silently “dramatic” defaults.** Ask for a growth chart and some assistants hand back a **tight y-axis** that flatters the trend: no warning, no comment. The chart runs; that’s what makes it dangerous.

...

### ! Important

Both threads, one rule: **an AI chart runs long before it’s true**. You verify the columns *and* the axis, every time.

## Your turn

### Open the exercise

Scan the QR or type the link:

[python.tobiasvlcek.com/notebooks/ex\\_09\\_e/](https://python.tobiasvlcek.com/notebooks/ex_09_e/)



Any questions? Let me know!

## Before Session X

uv, Zed and Mistral Vibe have been on your laptop since Session VII. The finale adds **git**. Two things before you arrive, about 15 minutes:

1. **GitHub account** from Session I. None yet? [github.com](https://github.com), tonight
2. **git + GitHub CLI (gh)**: install, log in once. [Git Basics](#)

...

### ! Important

**Do it in advance**, and bring the laptop. Session X opens with Checkpoint 5 (Sessions VIII-IX), then builds on git from the first minute. Install fights you? Ask in class or by e-mail.

## To the Lab

### A chart in a script

The lab is a script, and a script has no cell that could show a chart. So every chart goes into a **picture file**:

```
plt.figure()                # OPEN: a fresh canvas, as before
plt.plot(days, revenue)
plt.title("This week's revenue")
daily_chart = plt.gca()     # KEEP: the chart, for the lab's checks
plt.savefig("daily.png")    # SAVE: the picture, into python-labs
plt.close()                 # CLOSE: the canvas
```

...

Forget `plt.close()` and the next chart draws on top of this one: one canvas, two charts. To look at the picture, click `daily.png` in Zed's project panel, the file list on the left.

## After the break: the lab

- Download `lab_09_pitch.py` from [Episode 9: The Pitch Deck](#) into `python-labs`. `orders_messy.csv` is still there
- Nine tasks, six with a chart. No task names the kind of chart: **you pick it**. At the end the script puts all six into `pitch_deck.png`
- Tobi's deck is in the script too. **Two things in it are wrong without any task saying so**: one in his numbers, one in what he claims
- From table to chart is not on the slides: **ask your agent**, then **look at the picture**

...

### ! Important

Show me `pitch_deck.png` and the last block before you leave, for feedback as last week. Then two minutes of practice for Checkpoint 5, which is still handed in from the browser: open `ex_09_e`, **Cmd/Ctrl+S**, menu → Download → *Download Python code*, and find the file.

## Wrap-up

### Three things to remember

1. **Every chart, one frame.** Open with `plt.figure()`, close with `plt.gca()` (in a script: `savefig`, then `close`); label the axes and title it. A chart nobody can read argues nothing.
2. **Match the chart to the question.** Categories → bar, distribution → histogram, two numbers → scatter, time → line. The wrong chart is a lie; and “no pattern” is a real finding.
3. **Axis from zero, verified code.** Growth charts start at **0**, and every AI-drafted plot gets checked: do the columns exist, does the axis start where you claim?

...

### Note

**Season finale next: git and the project kickoff.** Install git and [gh](#) this week. Session X needs them right after the checkpoint.

## Literature

### Books to start with

- Wilke, C. (2019). Fundamentals of data visualization: A primer on making informative and compelling figures (First edition). O'Reilly Media. [Link to the free book website](#)
  - The best single book on *why* a chart works: principles over code. Highly recommended.
- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. [Link to free online version](#)

...

### Note

Building charts with AI this session? Revisit the [AI Tools page](#): describe-then-verify is the whole workflow. Matplotlib's own [pyplot tutorial](#) is a friendly next step.

...

For more, see the [literature list](#) of this course.